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Flame characteristics in a novel petal swirl burner

Lingling Zhao *, Qiangtai Zhou, Changsui Zhao

School of Energy and Environment, Southeast University, Nanjing 210-096, China

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Abstract

A three-dimensional (360°) body-fitted coordinate mathematical model to simulate pulverized coal particle combustion in a petal swirl burner (PSB) is first set up to analyze the flame stability and its characteristics. The studies on the flow pattern, the temperature distribution, and the flue gas composition of the flame, the ignition location, and the combustion efficiency of the pulverized coal particle are conducted. The results show that owing to the special geometric design of the PSB, some of the pulverized coal particles leaving the burner can directly enter the radial recirculation zone (RRZ) behind the petal flame stabilizer (PFS) and are immediately ignited and burned in the RRZ, producing a sort of flame that is always on duty behind each petal, which is called the permanent flame. The flame pattern, which is a combination of the main flame and several permanent flames, provides a sufficient heat source for reliable ignition and steady combustion even for the low-volatile coal-firing and turndown capacity operation, and is advantageous to lower NO_x emission. Moreover, the mechanisms by which the special flame pattern of PSB can be existed are analyzed. A PSB test was undertaken in a 210-MW power plant boiler to investigate the performance of the PSB with firing of low-volatile pulverized coal. The temperature measurement value along the burner axis is given, in which the temperature distribution and the ignition location are clearly shown.
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Keywords: Petal swirl burner; Flame stability; Numerical simulation; Flame pattern

1. Introduction

Electric power generation from power plants with pulverized coal-fired boilers is the major electricity source in China, and coal consumption for generating electricity has risen dramatically, with pollutant emissions being increased rapidly in the past 20 years. From the viewpoint of continuous and persistent development strategy, energy saving and environmental protection become the first important responsibility to be carried out [1,2]. Therefore, the development of a combustion system with stable combustion of coal, high efficiency of combustion, and minimum pollutant emissions from the flue gas of the boilers remains one of the active research issues in the field of coal combustion.

The existing state of coal applications in China shows that the use of low-volatile coal, such as anthracite and semi-anthracite coal (or lean coal), and low-rank coal (high-ash coal) for boiler fuel in the power plants is quite common [3,4], but the unstable combustion problem appears to be a vexing issue, even though the boiler burner is designed for the combustion of low-volatile or low-rank coals.

* Corresponding author.
E-mail address: zhao_lingling@seu.edu.cn (L. Zhao).

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Swirl burners with opposite-wall arrangement have been widely put to practical use for coal-fired boilers in power plants [5–7]. The main disadvantage of the swirl burners is unstable combustion while low-volatile coal is burned. For example, the phenomenon of flame extinction in a PAX-DRB type burner of a coal-fired boiler with capacity 210 MW in Xinhai Power Plant occurs frequently as low-volatile coal is burned. The flame extinction of a swirl burner with a bluff body stabilizer in a 210-MW boiler of Huangdao Power Plant burning low-volatile coal is another example, which occurs when the boiler load exceeds even 75–80% of rated capacity. Therefore, flame stability for the swirl burners is one of the basic factors to be taken into account in the development of a new type of swirl burner, and it is necessary to study the swirl flame pattern and to understand the influences on the mechanism of flame pattern formation.

The flame stability of coal combustion depends to a great extent on whether the ignition of coal particles is rapid or not. The ignition and combustion of the coal particles release thermal energy resulting from rapid chemical reaction, raise the recirculating flue-gas temperature to a high level, and quickly heat up the successive coal particles by means of intense mixing with the high-temperature recirculating flue-gas flow [8,9]. The high-temperature combustion flame is then sustained.

One way to achieve the rapid ignition of the coal particles is to create a high-temperature continuous flame right behind the burner. The stable flame can provide the thermal energy for the heating and ignition of the coal particles. There are two important factors that affect the pulverized coal ignition and the stable combustion for the swirl burners. The establishment of the internal center recirculation zone (CRZ) creates the opportunity for the high-temperature flue gas flowing backward from the combustion flame to form a hot-gas reverse flow and a heat source for the burner exit. The energy supply level is dependent on the recirculation zone parameters, such as its width and length, reverse flow rate, and temperature. The movement situations of the pulverized coal particles (the solid phase), including the solid phase concentration and the distribution and trajectory of the coal particles, will affect the heating rate and the ignition of the coal particles. If the coal particles leaving the burner move quickly toward the secondary-air side, the heating rate decreases, the ignition of the coal particles will be dramatically delayed, and the flame extinction even occurs for the burning of low-volatile coal.

The mechanisms of the flow field, the reverse flow characteristics, and the solid-phase coal particle movement have been studied by many researchers [10–12]. The formation of the CRZ, gas-phase flow fields, and the solid-phase distribution, concentration, and trajectory can be aerodynamically simulated [13–16], providing a convenient method of understanding the mechanisms of the combustion flame pattern for a swirl burner [17,18]. In addition, the experiments are employed to research different coal particle types of the optical particle concentration to the flame and these experiments play a role in the current burner designed so that the particle was kept as a dense perfect mixture; for a detailed description of that mechanism we refer to the literature [19–22]. The formation of dense-lean combustion structures is a result that can always be observed, together with the swirl flame in highly turbulent combustors. The appearance of these organized the air-particle flow distribution structure that was found for nonpremixed combustion systems and is also independent of the applied flame type (jet, bluff body, or ring-stabilized flames). However, even though mostly particles can enter the recirculation, sometimes the temperature of the recirculating flue gas at the front of the recirculation zone is not high to the coal ignition temperature because the long distance reverses movement and heat and mass transfer.

In the present paper the flow field, the particle trajectory, and the flame pattern are described by the three-dimensional model and observation. The main purpose of our research work is to create a stable high-temperature heat source near the burner exit; that is, there is a stable flame or heat source at the front of the main recirculation zone, to provide sufficient heat for igniting the successive pulverized-coal particles. The key point is to establish a way in which part of the coal particles from the pulverized coal stream can directly enter the recirculation zone and the main recirculation zone is not destroyed by the impingement of the pulverized-coal air stream. Therefore, the rapid ignition of the coal particles can be achieved. In this paper, the flow pattern, the coal particle movement, and the flame characteristics for a novel swirl burner are studied, and how to create a high-temperature flame right at the burner exit and what characteristics the flame has are also investigated.

2. Experimental burner

In order to research the effects of the flow characteristics on the ignition issue and the flame stability for the combustion of low-volatile and low-quantity coal in a swirl burner, according to the principles mentioned above, a new-type stabilizer of the swirl burner (petal flame stabilizer, PFS) is developed which has received a patent in China. The burner to which the new type stabilizer is applied is called a petal swirl burner or PSB for short, and this combustor is illustrated schematically in Fig. 1. The fuel

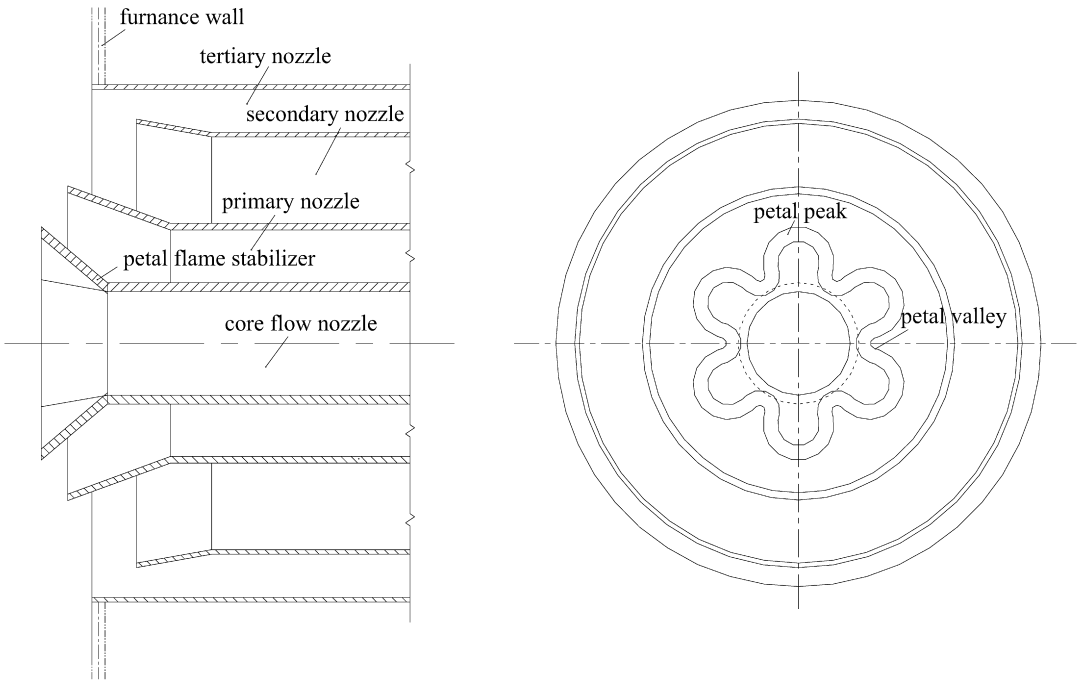


Fig. 1. Structure of petal swirl burner.

(pulverized coal) stream is introduced through the annular channel between the primary air nozzle and the central tube and the combustion air channel consists of the secondary nozzle and the tertiary nozzle. The combustion air swirling is maintained by a tangential-entry swirl generator located upstream of the burner throat that produced the solid-body rotation flow in the secondary nozzle. The original stabilizer is a surface-curved, gradually enlarged, and hollow cone, which is connected to the central tube (core tube) outlet end. The burner does not operate well and is characterized by high unburned carbon and frequent flame extinction when firing low-rank coal. The hollow cone stabilizer is replaced with a PFS in the experimental burner to form the PSB. From the back view the stabilizer appears to be a flower with several petals. The curved surface of the stabilizer consists of two parts, a convex surface and a concave surface. The outermost convex is given the name of the petal peak, and the innermost concave is called the petal valley.

The stabilizer is connected to the outlet end of the core for the swirl burner, which plays the role of a bluff body with the central air flow (flowing inside the core tube) being shut down. When the primary-air coal stream flows through the PSB, a strong central recirculation zone is formed behind the stabilizer; a large amount of high-temperature flue gas will flow backward to the burner exit to heat up the primary air–coal stream. The circumferential length of the contact boundary between the primary-air coal flow and the high-temperature recirculating flue gas is much larger than that of the burner, having a common bluff body or hollow cone. This means that the mixing area of the coal particles with the high-temperature flue gas is expanded more. The flow fields and characteristics of this type of stabilizer have been analyzed. As will be seen from the flow field of the PFS, there exists a secondary radial recirculation zone behind each of the petals besides the central recirculation zone (CRZ). Therefore, the mixed intensity of the recirculating flue gas flow with the primary air–coal stream is higher than that of the common swirl burner. In addition, moving annularly through the PFS of the burner, the primary-air coal stream is divided into two parts. The first part flows outward by the outward leading effect of the outermost convex (petal peak area), and the second part moves inward and along the flower valley. Some of the coal particles follow the air flow of the latter part and directly enter the central recirculation zone, in which rapid ignition of the coal particles occurs because of high temperature and less air (fuel-rich) conditions.

This special petal swirl burner (Fig. 1) is used in a semi-anthracite in a 210-MW power station boiler. A three-dimensional (360°) body-fitted coordinate mathematical model is first set up to simulate the pulverized coal particle combustion for the PSB. The flow pattern, the temperature distribution, and the flue gas composition of the flame are calculated. The ig-

Table 1
Most physical and chemical processes of coal combustion

Physical and chemical process	Sub-model	The selected model
Gas phase turbulent and recirculation	Gas turbulent model	RNG k–ε model
Pulverized coal dispersion	Two phase turbulent dispersion model	Discrete random walk model
Devolatilization model		Two-competing-model
	Coal combustion model	
Char combustion		Kinetics-diffusion model
Mixture model	Turbulent combustion model	PDF model

nition of the pulverized coal particles and the flame formation of the PSB are analyzed. The experimental measurements for the temperature field of the PSB installed in the boiler are taken and compared with the calculation results.

3. Mathematical model

The governing equations are solved numerically for the simulation of all processes, such as turbulent flow, coal combustion, solid particle transportation, mass transfer, and radiative and convective heat transfer. The main combustion submodels are listed in Table 1.

The gas flow is simulated with the Euler assumption, and since the flow is turbulent, the widely used RNG k–ε model [23,24] is coupled to close the turbulence problem. The coal combustion model comprises volatile yield, homogeneous combustion, and char heterogeneous oxidation. The devolatilization rate is simulated using two competing models [25], which implies that the rate of production of volatile gases is defined by two competing process. The homogeneous combustion of volatiles released from the particle is simulated using the mixed burnt model [26]. The instantaneous mass fractions are given in terms of the instantaneous mixture fraction. The mean mass fractions of fuel, oxidant, and combustion products are obtained from the mean and variance of the mixture fraction assuming the probability density function (PDF) [27,28]. The coal combustion model has to be combined with a particle transportation calculation. A Lagrangian approach has been chosen, considering the influence of a diluted particle phase on the fluid flow [29]. The interactions between particles have been neglected. The thermal radiation in the furnace is the dominant heat transfer mechanism due to the presence of a mixture of participative gases and particles at high temperature. The radiative heat transfer has been simulated using the discrete transfer method [13], which solves a transportation equation for the radiation intensity along the paths between two boundary walls. The influence of the particles, also participating in the radiative heat transfer, is taken account of using a specific heat source in the energy conservation equation.

Table 2
Composition and heating value for the case study coal

Proximate analysis (as received)	
Moisture (%)	7.12
Ash (%)	25.56
Volatile (%)	14.39
Fixed carbon (%)	52.93
Ultimate analysis (as received)	
Carbon (%)	58.62
Hydrogen (%)	3.02
Nitrogen (%)	0.97
Oxygen (%)	4.40
Sulphur (%)	0.31
Heating value (as received)	
Gross calorific value (kcal/kg)	5340

The calculation coal is a type of semi-anthracite with volatile 14.39% selected for the experimental test, whose ultimate and proximate analyses are listed in Table 2. High-volatile and high-quality coal can be ignited quickly and combust stably. In order to analyze the flame stability of PSB, we select this semi-anthracite coal.

4. Results and discussion

The three-dimensional computational fluid dynamic (CFD) models constitute a powerful tool to deal with the structure–complexity simulation of a PSB and to investigate the processes taking place in the boiler, providing a great number of precise numerical values for velocity, temperature, and concentration fields, irradiation profiles, heat transfer distribution, and pollutant emission. Simulations have been carried out concerning the predictions of the flame characteristics for different boiler loads. A complete simulation, including fluid flow, coal combustion, heat transfer, and particle trajectory, has been performed for each case to analyze the flame characteristics of this type of burner.

4.1. Flame pattern

Predictions of temperature distribution and flame pattern at 100% boiler load are shown in Figs. 2 and 3.

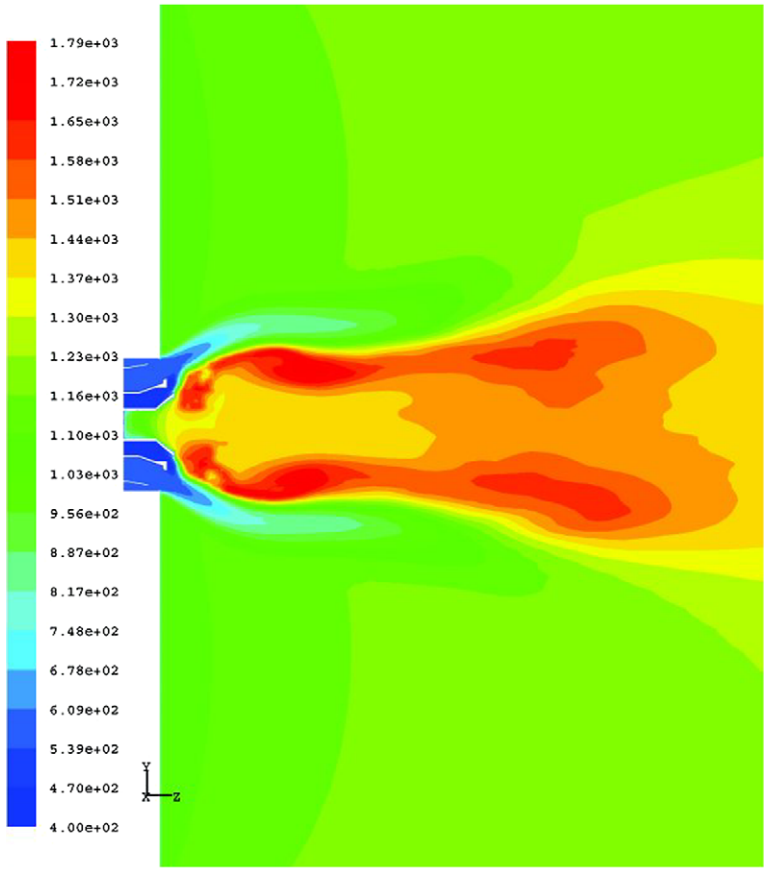


Fig. 2. Flame pattern of PSB at petal peak plane (K).

Fig. 2 describes the temperature distribution for the petal peak plane, while Fig. 3 indicates that of the petal valley plane. These figures show that the ignition occurs immediately behind the burner nozzle exit, with a complex flame pattern. With the special design of the PSB burner, the whole thick flame of the burner is split up into a thin annular and six (petal number) separate flames as seen in Fig. 4. The flame pattern is special, with separate flames at the front of the recirculation zone immediately behind the petals, and the main flame extended for meters. These separate flames are stable and can generate an adequate heat source to heat up and ignite the coal particles; therefore, the separate flame is called a "permanent flame." The pulverized coal particles entering the furnace through the petal valley area are heated by the heat source from the permanent flame on both sides and ignited rapidly. This flame pattern is profitable for the coal combustion.

Generally, near the petal peak surface, the temperature level is of great benefit to the coal combustion. When entering the furnace, the rapid ignition of the coal particle results in a high-flame-temperature region reaching about 1300–1500 °C at the permanent flame area. The flame at the recirculation boundary is also in the high-temperature region, and the flame has a long shape; the whole flame extends to a great distance. In the axial area of the burner, there is a lack of oxygen, so the flame temperature is not so high, roughly about 1000–1300 °C. The flame temperature is at the level of 1100–1500 °C as a whole. From the petal valley plane view, we can also see the existence of the permanent flame close behind the concave area. The highest temperature of the PSB is not in the center recirculation zone. For the PSB, the highest temperature zone is located in two regions. The first high-temperature region is at the permanent flame as mentioned above, and the second one is in the boundary area, where the intense mixing between the pulverized coal–air flow and the high-temperature recirculated flue gas flow takes place with high transfer rates of heat and mass accompanying it. The flame in this area extends for meters and is called the main flame.

Fig. 4 shows the combustion process of the PSB flame. The figure is taken at different distances from

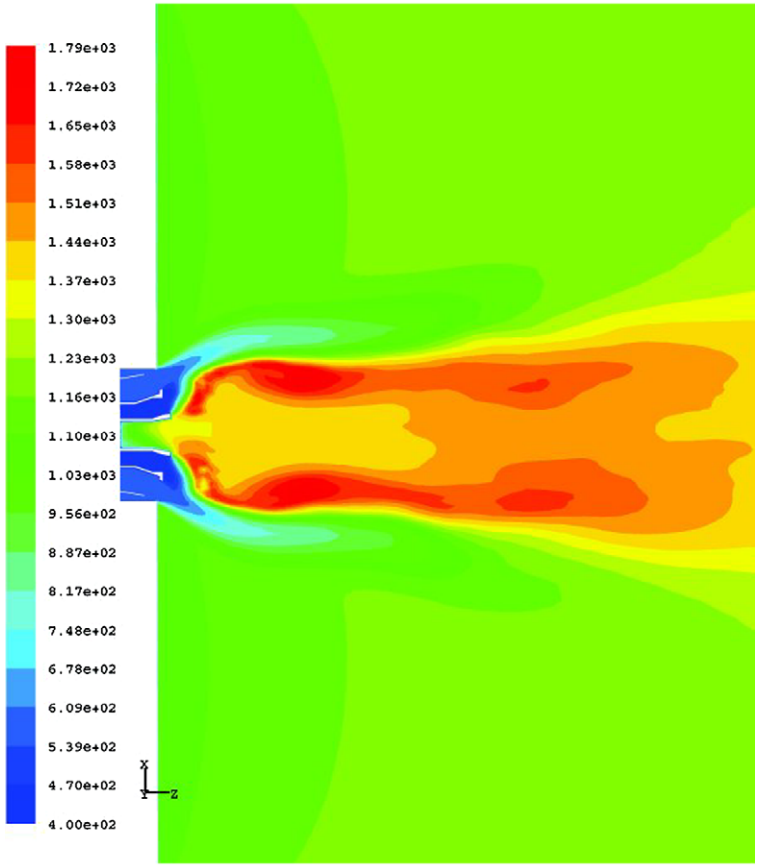


Fig. 3. Flame pattern of PSB at petal valley plane (K).

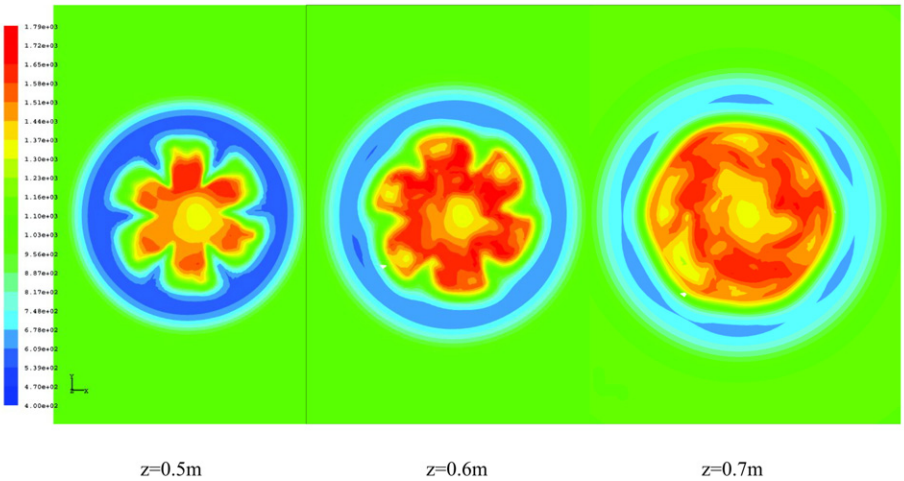


Fig. 4. Flame development process of petal swirl burner (K).

the burner exit: $z = 0.5$ m, $z = 0.6$ m, and $z = 0.9$ m. It is seen from this figure, that the front part of the flame also has the petal shape, and thus, the flame rotates with the swirl secondary flow. Finally, the petal-shaped flame is combined with the surroundings to form a round flame as a whole. The lower temperature zone in the figure is shown with a blue color, which corresponds to the unignited region of the coal

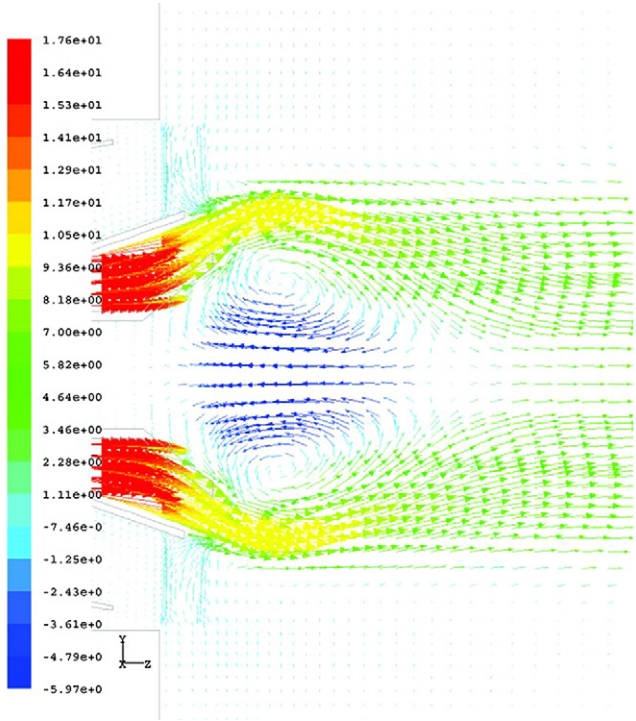


Fig. 5. Velocity vector field of stabilizer at petal peak surface (m/s).

particles and of the secondary air flow. In the radial recirculation zone, at the position $z = 0.5$ m, the flame temperature is as high as 1300–1500 °C; not far behind that position, the temperature is slightly higher. The high-temperature flame zone extends for meters at the boundary of the mixing area between the main flow and the recirculation flue gas flow.

4.2. Formation of the special flame

Why does the PSB form this special flame pattern? It corresponds with the flow pattern and particle trajectory of this type of burner. Figs. 5 and 6 show the velocity distribution of the stabilizer, through which only the primary air flow is introduced into the furnace. It can be seen that a vigorous radial recirculation zone (RRZ) is recognized. The flame pattern of the PFS is characterized with several sorts of recirculation zone. There exist a radial recirculation zone and a pair of axial recirculation zones behind each of the petals, besides the central recirculation zone (CRZ). The radial recirculation zone of the PFS plays an important role in the ignition of the pulverized coal particles. When the pulverized coal air stream flows through the PFS of the burner, part of the coal particles pass through the petal valley area into the recirculation zone. After changing direction and turning to the radial recirculation zone with outward direction first and downward direction later, this part of the particles are recirculated in the RRZ. The recirculating coal particles in the RRZ are situated in conditions of easy ignition because of high temperature, less excess air (fuel-rich condition), and more resident time. And therefore, the particles are heated up quickly, ignited rapidly, and burned steadily within the RRZ to form a high-temperature permanent flame, which generates a heat source to heat up the pulverized coal particles.

The particle trajectory of the PSB of the burner can also explain the existence of the permanent flame. Figs. 7a and 7b show the particle trajectories of the PSB flowing through the petal peak and petal valley areas, respectively. It is seen that the particle flowing through the petal valley area moves straight ahead and enters the recirculation zone with a stay of two or three turns in the RRZ. The particle across the petal peak area slightly diffuses outward before changing direction inward, and moves in the RRZ with a quick turn. This means that parts of the coal particles can directly or indirectly flow into the RRZ and recirculate in this region.

The existence of the RRZ, the characterization of the PSB flame, combined with the ordinary central recirculation zone, gives vital energies to the boundary between the pulverized coal flow and the high-temperature recirculating flue gas flow. The heat and mass transfer between them becomes more intense because the transfer takes place with not only the

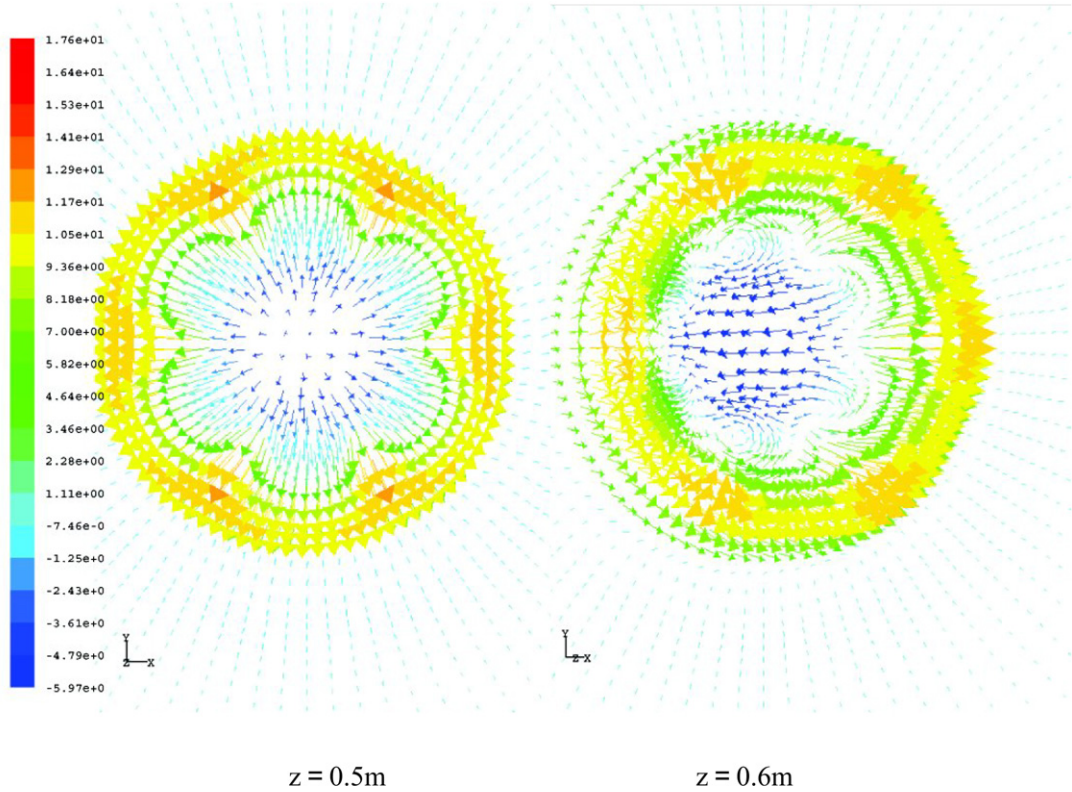


Fig. 6. Radial recirculation zone behind the stabilizer (m/s).

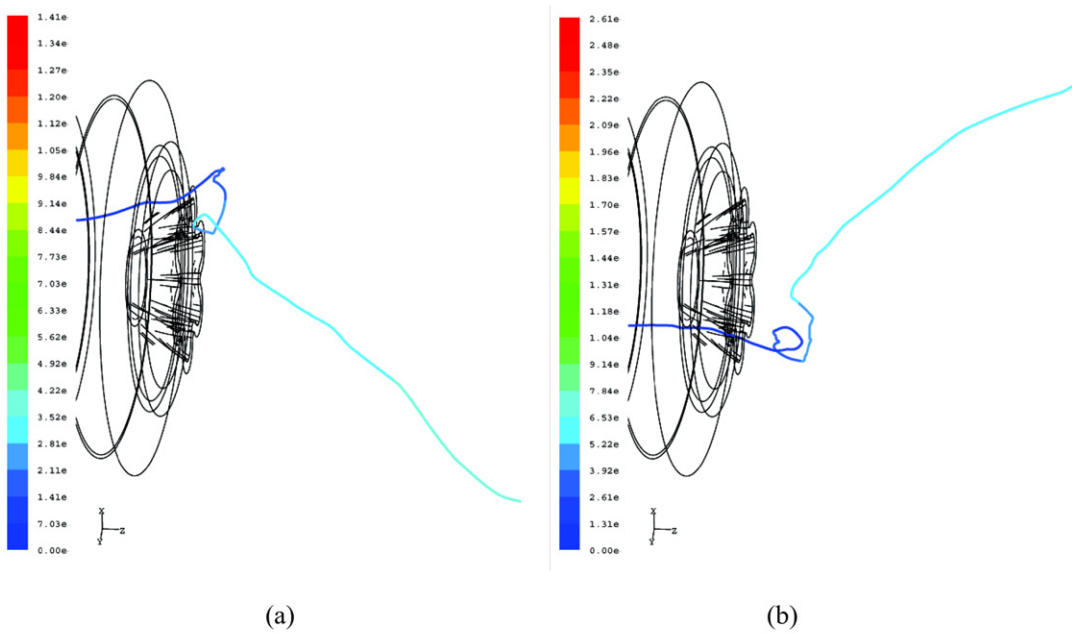


Fig. 7. Particle trajectory of petal swirl burner. The value is residence time (s). (a) is petal peak area; (b) is petal valley area.

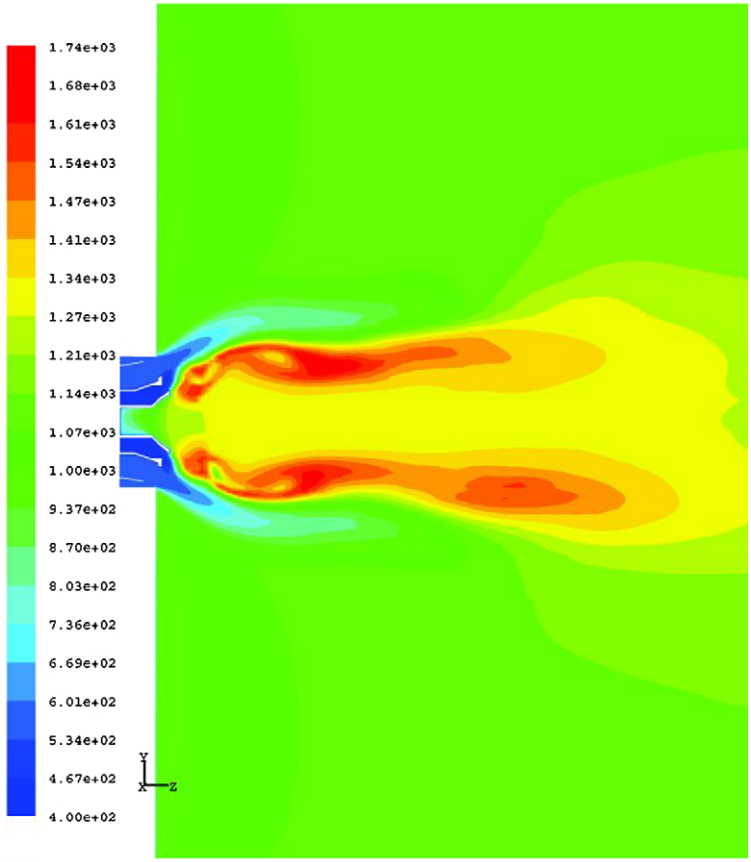


Fig. 8. Flame pattern of PSB at 75% rated capacity of boiler (K).

microscopic turbulent fluctuation that occurs at the boundary shear-layer of the recirculation zone in the ordinary swirl burners, but also the macroscopic intensive convection and mixing. The exchange rates of heat and mass transfer between the pulverized coal particles and the high-temperature recirculating flue gas at the boundary are enhanced, which accelerates the ignition and the combustion of pulverized coal particles. Therefore the second high-temperature region near the boundary is formed, which is called the main flame.

It is clear that the PSB has many important features, including the formation of a multizone of the recirculating flue gas and the direct entry of the coal particles into the recirculation zone. These features are propitious for the stable combustion that is especially important with low-volatile and low-rank coals. The pulverized coal particles entering the recirculation zone are burned under less excess air (fuel-rich) conditions because of the large quantity of coal particles that undergo rapid ignition and exhaust oxygen quickly, which is advantageous to the attenuation of the NO_x emission.

4.3. *Flame temperature for low capacity*

The temperature fields of the flame at boiler load of 75 and 55% of rated capacity, respectively, are shown in Figs. 8 and 9 for comparison. As can be seen from Fig. 8 for 75% of rated capacity, the size of the high-temperature region for the main flame has decreased, but the temperature level is roughly the same as in Figs. 2 and 3. For the boiler load of 55% of rated capacity, the high-temperature region of the main flame decreases further, and the value of the highest temperature of the flame decreases somewhat. The permanent flame is still existent, with a temperature level of 1300–1400 °C, even turning down the boiler load to the value of 55% of rated capacity, as seen in Fig. 9.

5. The measured data

The furnace measured data used in this paper originate from an industry experiment carried out in the Huangdao Power Station 210-MW boiler. There is an

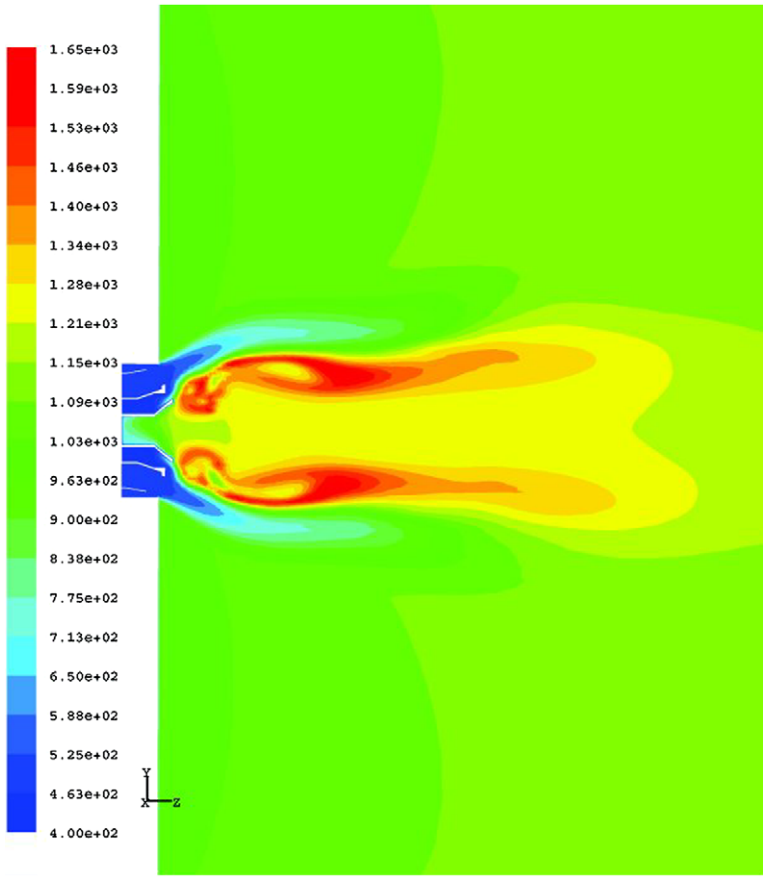


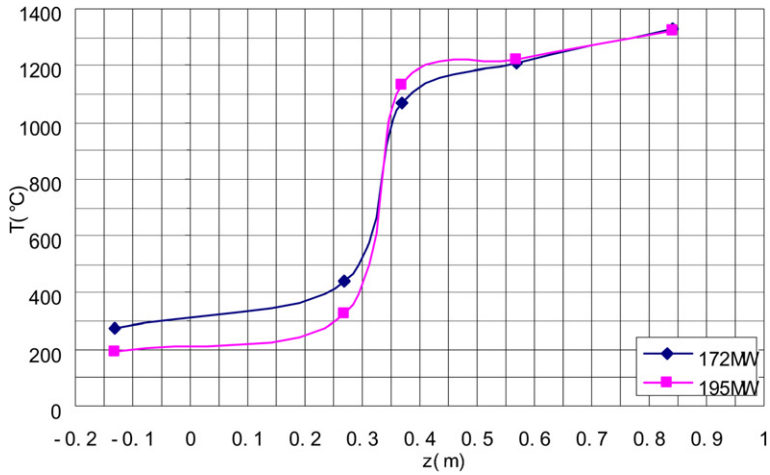
Fig. 9. Flame pattern of PSB at 55% rated capacity of boiler (K).

inspection hole fort each burner on the center of the core pipe, and through it we can see the flames and measure the temperature along the center line for different coal and boiler loads.

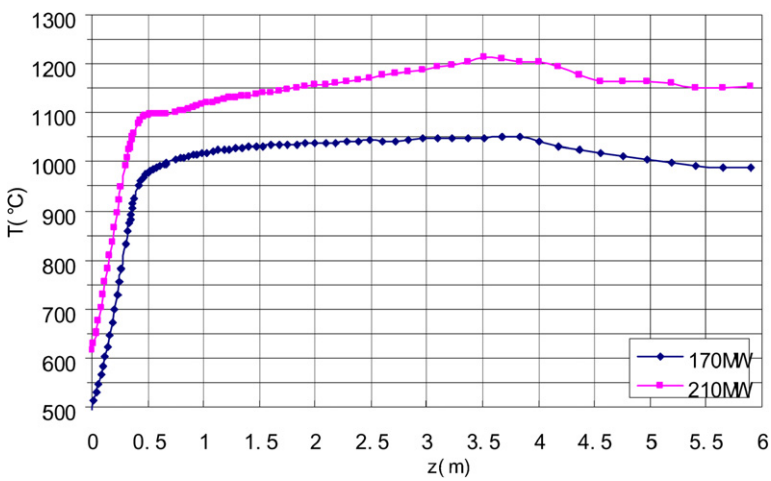
Observing through the central inspection hole of the burner, we can see the coal particle stream flowing, the particles entering the recirculation zone, the particles changing direction, and the successive flashes of the coal particles. We find that the flame is not outstandingly bright on the burner axis. This proves the movement of the coal particle into the recirculation. The adjustment of the secondary air baffle does not influence the stable combustion of the PFB burner. The results of the industry experimentation indicate that the PSB can stably burn the semi-anthracite coal with Vdaf = 12–18% in 55–100% boiler load. Fig. 10a is the experimental data of the flame temperature for the centerline of the burner measured from the inspection hole at the center of each burner. The solid line in this figure is the experimental results for different operation conditions, which shows that the particle can ignite within 200 mm of the burner exit. Fig. 10b shows the numerical simulation data of the centerline at corresponding boiler loads. The comparison of Fig. 10 shows that the flame development tendency, especially for the coal ignition stage, is fairly similar between the experiment and the numerical simulation.

6. Conclusions

The design of the structure of the PFB burner has taken the mixing of the primary air and pulverized coal flow with the recirculation flue gas carefully into consideration. The pulverized coal particles can enter the recirculation zone, and then mix rapidly with the recirculating flue gas because of the great intensity of the macroscopic convection transfer. The permanent flames provide a stable and sufficient heat source for the pulverized coal ignition. This special flame pattern is profitable for reliable ignition and burnoff of the low-volatile pulverized coal, for turndown capabilities, and for attenuation of the NO_x emission.



(a)



(b)

Fig. 10. Comparison of (a) measured and (b) calculated temperature values along the axis.

Acknowledgments

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